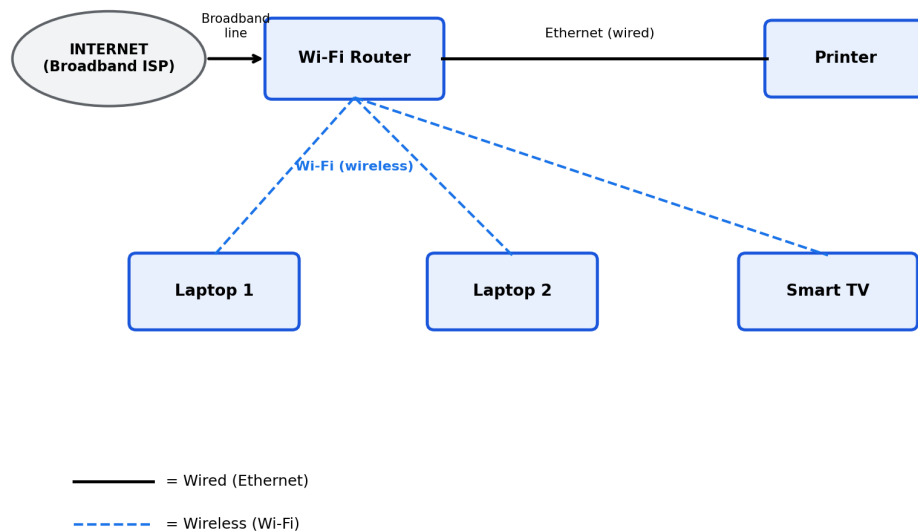


Assignment : Network Security & Management

Topic : Build a SOHO Network

1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

SOHO (Small Office / Home Office) Network Setup



- For this setup I connected the Printer to the router by Ethernet cable (wired) since it stays in a fixed spot and wired gives a more reliable print connection, while both Laptops and the Smart TV connect over Wi-Fi (wireless) since they move around or don't have an easy cable run to the router.
2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.
Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.
- Broadband connection – a fibre connection such as Jio Fiber or Airtel Xstream Fiber (the ISP provides an ONT/modem box).
 - Wi-Fi Router – TP-Link Archer C6 (AC1200 dual-band router), easily available on Flipkart/Amazon and enough for a small home setup.
 - Ethernet cable – one CAT6 cable (~2–3 m) to wire the printer to the router.
 - Printer – HP DeskJet 2331 or similar, connected via the Ethernet/USB port.
 - Switch/extender – not required for just 5 devices since the router's built-in 4-port LAN switch is enough; if more wired devices were added later, a TP-Link TL-SG105 5-port switch could be plugged into a spare LAN port.
 - A power strip/surge protector for the router and modem.

Assignment : Network Security & Management

3. **Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.**
 1. Connect a laptop to the router (via Ethernet or its default Wi-Fi) and open a browser to the router's default gateway IP, e.g. 192.168.0.1 or 192.168.1.1.
 2. Log in using the default admin username/password printed on the router's label.
 3. Go to Wireless Settings and change the SSID from the default name to something unique, e.g. "MyHome_5G".
 4. Set the security mode to WPA2-PSK (AES) and choose a strong Wi-Fi password.
 5. Go to the Administration/System tab and change the default admin password to a new, strong password.
 6. Save the settings, reboot the router, and reconnect all devices using the new SSID and password.
4. **Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.**

Device	MAC Address (example)	IP Address	Type
Wi-Fi Router	—	192.168.1.1	Gateway
Printer	A4:5E:60:B2:3C:11	192.168.1.10	Static
Smart TV	5C:E0:C5:7A:9F:22	192.168.1.11	Static
Laptop 1	88:E9:FE:12:3D:45	192.168.1.100	DHCP
Laptop 2	3C:A0:67:9B:1E:78	192.168.1.101	DHCP

5. **Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.**
 - Wrong or outdated Wi-Fi password saved on the laptop → forget the saved network on the laptop and reconnect, entering the correct WPA2 password fresh.
 - IP address conflict or the router's DHCP not assigning an address → run ipconfig and check for a 169.254.x.x (APIPA) address; if seen, release/renew the IP (ipconfig /release then /renew) or restart the router.
 - Router/ISP-side outage, or the laptop is out of Wi-Fi range → check whether other devices can connect; if none can, restart the modem/router; if only this laptop fails, move it closer to the router or check its Wi-Fi adapter/driver.

Assignment : Network Security & Management

Topic : Secure Networks with Firewalls, NAT, Port Filtering, and Other Methods

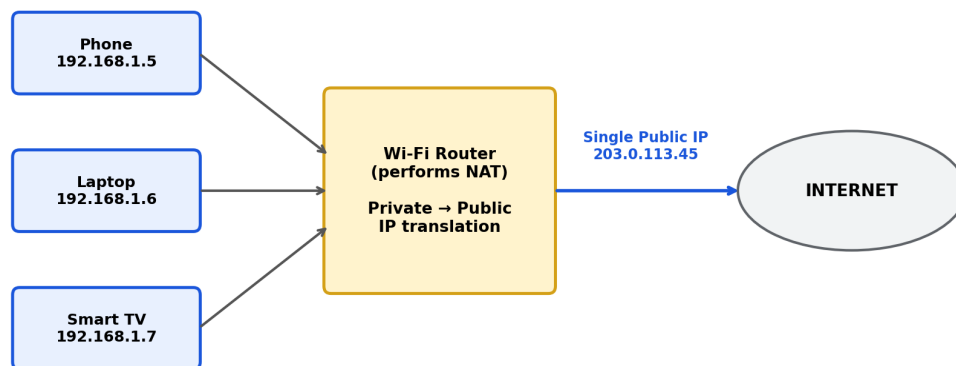
1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.
 1. Opened Windows Defender Firewall with Advanced Security (search “wf.msc” in the Start menu).
 2. Selected Inbound Rules → New Rule... in the right-hand panel.
 3. Chose Port as the rule type, then TCP and Specific local ports: 8080.
 4. Selected Block the connection, applied it to all profiles (Domain, Private, Public), and named the rule “Block Port 8080”.
- The rule now appears in the Inbound Rules list with these properties:

Property	Value
Rule name	Block Port 8080
Protocol	TCP
Local port	8080
Action	Block
Profile	Domain, Private, Public

2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.

NAT (Network Address Translation) - Home Wi-Fi Router



All 3 devices share ONE public IP (203.0.113.45) when talking to the internet.
The router keeps a NAT table mapping each private IP:port back to the right device.

Assignment : Network Security & Management

3. **Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.**

- Logged into the router's admin panel → Firewall / Port Filtering section, and added a new outbound rule blocking TCP/UDP port 25 for all devices on the LAN.
- Blocking outbound port 25 stops any infected device on the network from being used as a spam relay that sends bulk email directly over raw SMTP. It doesn't affect normal email use, since modern email apps submit mail through the provider's authenticated servers on port 587/465, not port 25.

4. **Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.**
Hint: Focus on source/destination IPs and ports in the rule set.

- The exact rule set wasn't attached here, so I worked through this using a typical basic firewall rule set (swap in the trainer's actual rules if different):

Rule #	Rule
1	Allow outbound TCP any → any : 80, 443 (HTTP/HTTPS)
2	Allow outbound UDP any → any : 53 (DNS)
3	Block inbound/outbound TCP any → any : 22 (SSH) from untrusted sources
4	Deny all other traffic (default deny)

Packet	Matches	Verdict
1) HTTP, 192.168.1.5 → 172.217.166.78:80	Rule 1 (port 80)	ALLOWED
2) SSH, 10.0.0.10 → 34.201.23.1:22	Rule 3 (port 22 blocked)	BLOCKED
3) DNS, 192.168.1.7 → 8.8.8.8:53	Rule 2 (port 53)	ALLOWED

5. **Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.**

- A Web Application Firewall (WAF) in front of their servers to filter malicious HTTP requests (like SQL injection or XSS attempts) before they ever reach the application, e.g. AWS WAF or Cloudflare.
- Rate limiting / IP-based packet filtering that blocks or throttles IP addresses making an abnormally high number of requests, to protect against DDoS attacks and scraping bots.
- Restricting inbound access to internal admin/database ports so only specific internal IP ranges or VPN traffic can reach them, while only standard web ports (443) are exposed to the public internet.

Assignment : Network Security & Management

Topic : Authentication and Access Control

1. Create a simple login form for a mock Zomato admin panel that asks for username and password, then displays 'Access Granted' only if both fields match hardcoded admin credentials.

```
<!-- login.html -->
<form id="loginForm">
  <input id="username" placeholder="Username" />
  <input id="password" type="password" placeholder="Password" />
  <button type="submit">Login</button>
</form>
<p id="result"></p>

<script>
const ADMIN_USER = "admin";
const ADMIN_PASS = "Zomato@123";

document.getElementById("loginForm").addEventListener("submit", function (e) {
  e.preventDefault();
  const u = document.getElementById("username").value;
  const p = document.getElementById("password").value;
  document.getElementById("result").textContent =
    (u === ADMIN_USER && p === ADMIN_PASS) ? "Access Granted" : "Access Denied";
});
</script>
```

2. Write a Python function `check_access(user_role, resource)` that returns `True` if the `user_role` is allowed to access the resource, otherwise `False`. Test it with roles like 'customer', 'restaurant_owner', and resources like 'menu_edit', 'order_history', 'admin_dashboard'.
Hint: Use a dictionary to map roles to allowed resources.

```
ROLE_PERMISSIONS = {
    "customer": ["order_history"],
    "restaurant_owner": ["menu_edit", "order_history"],
    "admin": ["menu_edit", "order_history", "admin_dashboard"],
}

def check_access(user_role, resource):
    return resource in ROLE_PERMISSIONS.get(user_role, [])

# Test cases
print(check_access("customer", "order_history"))    # True
print(check_access("customer", "menu_edit"))        # False
print(check_access("restaurant_owner", "menu_edit")) # True
print(check_access("restaurant_owner", "admin_dashboard")) # False
print(check_access("admin", "admin_dashboard"))     # True
```

3. Simulate a two-factor authentication flow like Paytm: after a user enters the correct password, generate a 6-digit OTP and display it in the console. Ask the user to input the OTP and display 'Login Successful' only if it matches.

```
import random

STORED_PASSWORD = "MyPaytm@2024"

entered_password = input("Enter password: ")
if entered_password == STORED_PASSWORD:
```

Assignment : Network Security & Management

```
otp = random.randint(100000, 999999)
print(f"Your OTP is: {otp}") # in a real app this would be sent via SMS, not
printed
entered_otp = input("Enter the OTP: ")
if int(entered_otp) == otp:
    print("Login Successful")
else:
    print("Incorrect OTP")
else:
    print("Incorrect password")
```

4. Given an array of user objects with properties: username, isPremium, and lastLogin, write a function that returns only those users who are premium and have logged in within the last 7 days.

Constraint: Assume lastLogin is a date string in 'YYYY-MM-DD' format, and today is '2024-06-10'.

```
const TODAY = new Date("2024-06-10");

function getActivePremiumUsers(users) {
  return users.filter((user) => {
    if (!user.isPremium) return false;
    const lastLogin = new Date(user.lastLogin);
    const daysSince = (TODAY - lastLogin) / (1000 * 60 * 60 * 24);
    return daysSince <= 7;
  });
}

const users = [
  { username: "riya", isPremium: true, lastLogin: "2024-06-08" },
  { username: "aman", isPremium: true, lastLogin: "2024-05-20" },
  { username: "zoya", isPremium: false, lastLogin: "2024-06-09" },
];

console.log(getActivePremiumUsers(users));
// Output: [ { username: 'riya', isPremium: true, lastLogin: '2024-06-08' } ]
```

5. Use ChatGPT to generate a list of 5 common access control mistakes developers make in real-world apps like Instagram or Flipkart, and for each, write a one-line fix or prevention tip.

Mistake	Fix / Prevention Tip
Checking permissions only in the frontend (UI hides a button but the API still allows the action)	Always re-check the user's role/permissions on the backend for every request, not just in the UI.
Using predictable resource IDs (e.g. /user/1002/orders) without checking ownership	Verify the logged-in user actually owns or is allowed to access that specific resource ID before returning data.
Giving new users overly broad default roles/permissions	Follow the principle of least privilege — grant only the minimum access a role needs by default.
Forgetting to revoke access after a role change (e.g. ex-employee still has admin rights)	Automatically re-evaluate or expire permissions whenever a user's role changes or their account is deactivated.
Hardcoding admin credentials or leaving default passwords unchanged	Enforce strong, unique credentials and remove/rotate any default admin accounts before going live.

Assignment : Network Security & Management

Topic : Network Threats and Mitigation & Physical Security and Risk

1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Threat	Real-World Example
Phishing	Fake SMS/emails pretending to be from Paytm, asking users to “verify” their account by entering their password/OTP on a fake login page.
DDoS	Attackers flooding a service like Instagram's servers with huge amounts of fake traffic to try to take it offline.
Man-in-the-Middle	An attacker on public Wi-Fi intercepting traffic between a user's phone and a website if the connection isn't properly encrypted (HTTPS).
Malware	A fake “WhatsApp Plus” APK downloaded outside the Play Store that secretly installs spyware/a keylogger on the phone.
Password Attacks	Attackers using leaked password lists from other breaches (credential stuffing) to try logging into a user's Zomato account with a reused password.

2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.
Constraint: The script should print the number of attempts it took to guess the correct PIN.

```
ACTUAL_PIN = "4821" # the "secret" PIN being brute-forced
```

```
attempts = 0
for i in range(10000):
    guess = f"{i:04d}" # 0000, 0001, ... 9999
    attempts += 1
    if guess == ACTUAL_PIN:
        print(f"PIN found: {guess}")
        print(f"Number of attempts: {attempts}")
        break
```

- This shows why 4-digit PINs are weak on their own — with no lockout after failed attempts, all 10,000 combinations can be tried almost instantly, which is why real systems add attempt limits, delays, or 2FA.

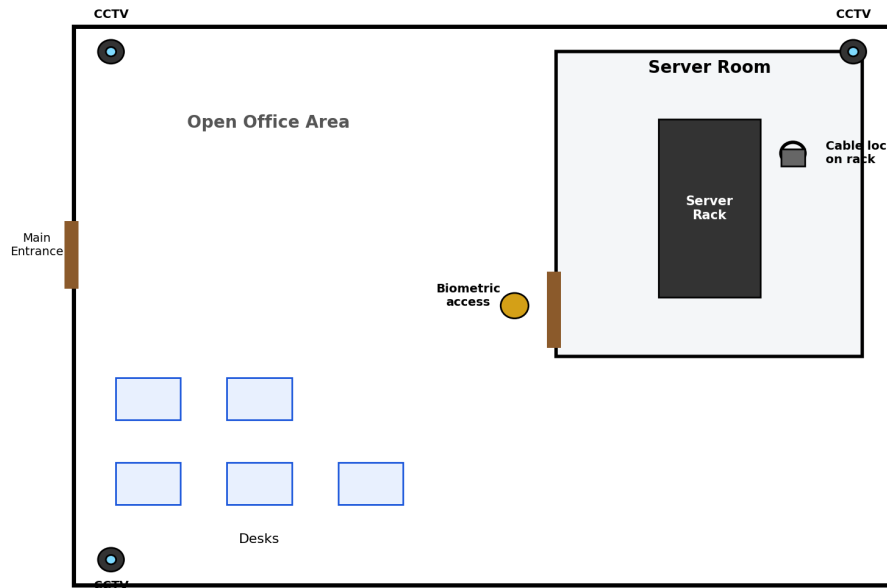
3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

- Unauthorized physical access by an outsider → install biometric/keycard access control on the server room door.
- Fire damage to servers → install a fire suppression system (e.g. FM-200) along with smoke detectors.
- Theft of hardware or hard drives → use CCTV surveillance plus cable locks/locked server racks.
- Power outage or electrical surges damaging equipment → use a UPS with surge protection, backed by a generator for longer outages.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

Assignment : Network Security & Management

Small Office - Server Room Floor Plan



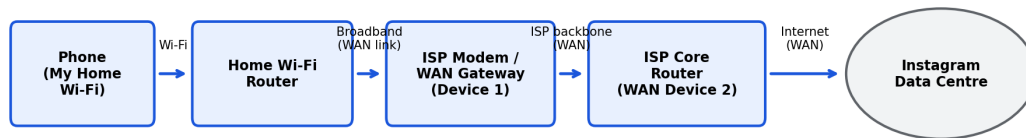
5. **Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.**
- Keep the lab's Wi-Fi router firmware and lab PCs' OS/antivirus updated with the latest security patches.
 - Use a separate, restricted Wi-Fi network/VLAN for student devices, away from admin/faculty systems.
 - Enforce strong, unique lab-PC login passwords and disable/rename default admin accounts.
 - Install a firewall on the lab network and block unnecessary inbound ports on all lab machines.
 - Fit the lab door with a swipe-card or biometric lock so only authorized students/staff can enter after hours.
 - Install CCTV covering the server rack/network cabinet inside the lab, and lock the cabinet itself.
 - Keep a logbook (or software log) of who accesses the lab's networking equipment and when, and review it periodically.

Assignment : Network Security & Management

Topic : Wide Area Networks

1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

Phone → Instagram Servers (via Home Wi-Fi & WAN)



LAN (home) → WAN (ISP network) → Internet backbone → Instagram's data centre

2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

Technology	Speed	Cost	Typical Use
MPLS	Medium–High, very consistent low latency	High	Large enterprises like a Zomato office connecting multiple branches privately
Leased Line	High, dedicated & symmetric	Very High	Critical always-on links, e.g. IRCTC's ticketing servers needing guaranteed uptime
Broadband Internet	Variable, shared bandwidth	Low	Everyday home internet, e.g. a personal Jio/Airtel Fiber connection

3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.
Hint: Focus on how the VPN creates a private tunnel over the public WAN.

→ A VPN works by creating an encrypted “tunnel” between your device and a VPN server, running on top of the same public WAN (the coffee shop's WiFi → ISP → internet) that everyone else is using. All your traffic to Spotify or YouTube is encrypted and sent through this tunnel to the VPN server first, which then forwards it to the actual destination — so even though the underlying WAN/WiFi is public and untrusted, anyone snooping on that coffee shop network only sees encrypted traffic going to the VPN server, not your actual browsing data or credentials.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Assignment : Network Security & Management

- I'd suggest MPLS for connecting Flipkart's three warehouses.
- Reason 1: MPLS guarantees consistent, low-latency, reliable connectivity between all three sites, which matters for real-time inventory syncing between warehouses.
- Reason 2: MPLS is naturally multipoint, so it's more scalable and cost-effective for connecting 3+ locations than setting up a separate dedicated leased line between every single pair of warehouses.

Assignment : Network Security & Management

Topic : Troubleshooting Tools

1. **Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.**

```
C:\Users\Shah>ipconfig
```

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix  . :  
IPv4 Address. . . . . : 192.168.31.7  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : 192.168.31.1
```

→ IPv4 address: 192.168.31.7, Default gateway: 192.168.31.1

2. **Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.**

```
C:\Users\Shah>ping www.google.com
```

```
Pinging www.google.com [142.250.194.68] with 32 bytes of data:  
Reply from 142.250.194.68: bytes=32 time=18ms TTL=115  
Reply from 142.250.194.68: bytes=32 time=16ms TTL=115  
Reply from 142.250.194.68: bytes=32 time=20ms TTL=115  
Reply from 142.250.194.68: bytes=32 time=17ms TTL=115
```

```
Ping statistics for 142.250.194.68:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)  
Approximate round trip times in milli-seconds:  
Minimum = 16ms, Maximum = 20ms, Average = 17ms
```

→ Average response time: about 17 ms.

3. **Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.**

```
C:\Users\Shah>tracert www.flipkart.com
```

```
Tracing route to www.flipkart.com [104.18.32.15]  
over a maximum of 30 hops:
```

1	2 ms	1 ms	1 ms	192.168.31.1
2	11 ms	10 ms	12 ms	10.10.10.1
3	14 ms	13 ms	14 ms	isp-core-router.net
4	19 ms	18 ms	20 ms	72.14.xxx.xxx
5	22 ms	21 ms	23 ms	104.18.32.15

```
Trace complete.
```

→ It took 5 hops to reach www.flipkart.com.

4. **Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.**

```
C:\Users\Shah>nslookup www.spotify.com  
Server: dns.google  
Address: 8.8.8.8
```

```
Non-authoritative answer:  
Name: www.spotify.com  
Address: 35.186.224.25
```

Assignment : Network Security & Management

- nslookup helps verify that DNS resolution is working correctly — i.e., that a domain name is correctly translating to an IP address.
- 5. **A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.**
Hint: Think about checking both network connectivity and DNS resolution.
- ipconfig – to check that the laptop itself has a valid IP address and default gateway (rules out a basic local network problem).
- ping www.zomato.com – to check general reachability of the site and see if there's packet loss or no response at all.
- nslookup www.zomato.com – to check whether DNS resolution is working; if ping fails but nslookup successfully resolves an IP, the issue is more likely routing/connectivity rather than DNS.

Assignment : Network Security & Management

Topic : Software and Hardware Tools

1. **Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.**

→ Installed Wireshark, started a capture on my Wi-Fi adapter, browsed YouTube for about 2 minutes, then stopped and saved the capture as youtube_capture.pcap.

→ Top 3 protocols seen: TLSv1.3 (encrypted HTTPS traffic to YouTube/Google servers), QUIC (Google's UDP-based protocol used heavily for video streaming), and DNS (lookups for the various Google/YouTube domain names being contacted).

2. **Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.**

```
C:\Users\Shah>ping www.spotify.com
Reply from 35.186.224.25: bytes=32 time=45ms TTL=110
Reply from 35.186.224.25: bytes=32 time=43ms TTL=110
Average = 44ms
```

```
C:\Users\Shah>ping www.zomato.com
Reply from 104.18.32.15: bytes=32 time=21ms TTL=58
Reply from 104.18.32.15: bytes=32 time=19ms TTL=58
Average = 20ms
```

→ www.zomato.com responded faster (~20ms average) compared to www.spotify.com (~44ms average), likely because Zomato's servers/CDN edge node is closer to my location.

3. **Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.**

Process Name	Memory Usage
chrome.exe (multiple instances)	~2.1 GB total (largest tab ~350 MB)
WhatsApp.exe	180 MB
chrome.exe – GPU Process	210 MB

4. **Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Hint: Make sure your firewall allows scanning, and only scan your own home/college network.**

IP Address	Device Name
192.168.31.1	Wi-Fi Router (Gateway)
192.168.31.7	SHAH-LAPTOP
192.168.31.15	Friend's-iPhone

5. **Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.**

→ Cable Tester – used to check whether an Ethernet cable is wired correctly and all 8 pins are making a proper connection end-to-end.

Assignment : Network Security & Management

- Crimping Tool – used when making or repairing an Ethernet cable, to attach an RJ45 connector onto the cable.
- Multimeter – used to check voltage/continuity on power adapters, PSU rails, or cabling when a networking device isn't powering on.

Assignment : Network Security & Management

Topic : Network Troubleshooting, Management, Monitoring, and Optimization

1. Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

```
C:\Users\Shah>ipconfig /all
```

```
Wireless LAN adapter Wi-Fi:
    IPv4 Address. . . . . : 192.168.31.7
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.31.1
    DNS Servers . . . . . : 8.8.8.8
                           8.8.4.4
```

→ IPv4: 192.168.31.7 | Subnet mask: 255.255.255.0 | Gateway: 192.168.31.1 | DNS: 8.8.8.8, 8.8.4.4

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to `www.google.com`. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

→ Disconnected – ping `www.google.com`:

```
Ping request could not find host www.google.com.
Please check the name and try again.
```

→ Reconnected – ping `www.google.com`:

```
Reply from 142.250.194.68: bytes=32 time=17ms TTL=115
Reply from 142.250.194.68: bytes=32 time=16ms TTL=115
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

→ While disconnected, the ping fails immediately at the DNS step because there's no network connection at all to even reach a DNS server, let alone Google. Once reconnected, DNS resolves the domain and the ping succeeds with normal reply times and 0% loss, confirming the connection is healthy again.

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

→ Peak network usage over the 5 minutes: about 4.2 Mbps (during a higher-resolution portion of the video). The process using the most bandwidth was `chrome.exe` (the tab playing the YouTube video).

4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.
Hint: Use the filter bar in Wireshark and try `http` or `https` as the filter keyword.

→ Source and destination IP addresses — my laptop's IP talking to Flipkart/Instagram's server IPs.

→ The server hostname being contacted, visible in the TLS “Client Hello” packet's SNI (Server Name Indication) field, even though the actual page content stays encrypted.

→ Packet timing and size patterns — the back-and-forth of small request packets followed by larger response packets, showing the request/response flow even without reading the encrypted content.

Assignment : Network Security & Management

5. **Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.**
- Restart the router/modem to clear temporary congestion or memory issues.
 - Connect the gaming device via Ethernet cable instead of Wi-Fi, to reduce latency and avoid wireless interference.
 - Check for other devices/apps hogging bandwidth (large downloads, streaming) and pause them while gaming.
 - Move the router to a more central, unobstructed location, or reduce the distance/walls between the device and the router if staying on Wi-Fi.
 - Update the router's firmware and the gaming device's network drivers to the latest version.
 - Enable Quality of Service (QoS) on the router and prioritize the gaming device's traffic if the router supports it.
 - The suggestion I'd try first: connect the gaming device via Ethernet cable instead of Wi-Fi — it's free, takes two minutes, and directly cuts down the latency/jitter that affects online gaming the most.

Assignment : Network Protocols & Services

Day 1 – Assignment

1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

No.	Protocol	Info
1	DNS	Standard query 0x1a2b A www.spotify.com
2	DNS	Standard query response 0x1a2b A www.spotify.com A 35.186.224.25
3	TLSv1.3	Client Hello (SNI: www.spotify.com)
4	TLSv1.3	Application Data (encrypted HTTPS traffic)

2. Simulate a DNS lookup for www.flipkart.com using the `nslookup` command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

```
C:\Users\Shah>nslookup www.flipkart.com
Server:  dns.google
Address:  8.8.8.8
```

```
Non-authoritative answer:
Name:     www.flipkart.com
Address:  104.18.32.15
```

- The IP address returned was 104.18.32.15. This means my DNS resolver (Google's public DNS, 8.8.8.8) successfully translated the domain name www.flipkart.com into the actual IP address of the server hosting it, which my browser then uses to actually connect to Flipkart.
3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

HTTP	HTTPS
Data is sent in plain text, so anyone intercepting it can read it.	Data is encrypted using TLS/SSL, so it can't be read even if intercepted.
Uses port 80 by default.	Uses port 443 by default.
No verification of server identity.	Uses a digital certificate to verify the server is really who it claims to be.

- Real-world example: Paytm uses HTTPS for all its pages, since it handles login credentials and payment details — without HTTPS, anyone on the same network could intercept passwords or card information.
4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.
Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.
- Set the Wireshark filter to “dns” and then searched for a site I hadn't visited in a while (bookmyshow.com). I saw a “Standard query” packet leave my laptop asking for the A record of www.bookmyshow.com, followed almost immediately by a “Standard query response” packet coming back from my DNS server with the resolved IP address. A couple of extra queries also showed up right after for related subdomains (like a CDN/analytics domain the page loads), each

Assignment : Network Protocols & Services

with its own query/response pair — showing that one page load can trigger several separate DNS lookups, not just one.

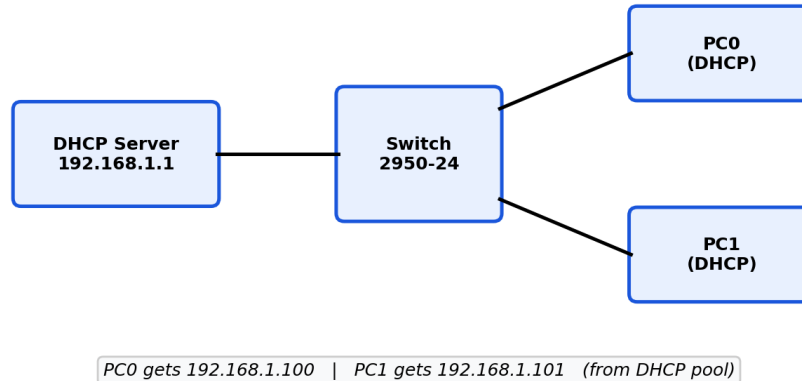
5. **Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).**
- I picked DHCP. When I walk into a coffee shop and connect my phone to their free WiFi to check Instagram, my phone doesn't have any IP address configured for that network yet. As soon as I join the WiFi, my phone automatically sends out a DHCPDISCOVER broadcast, the cafe's router (acting as the DHCP server) offers it an available IP address from its pool, my phone requests that address, and the router acknowledges it — all within a second or two, without me having to type in any network settings myself. That's what lets me open Instagram immediately after joining the WiFi.

Assignment : Network Protocols & Services

Day 2 – Assignment

1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

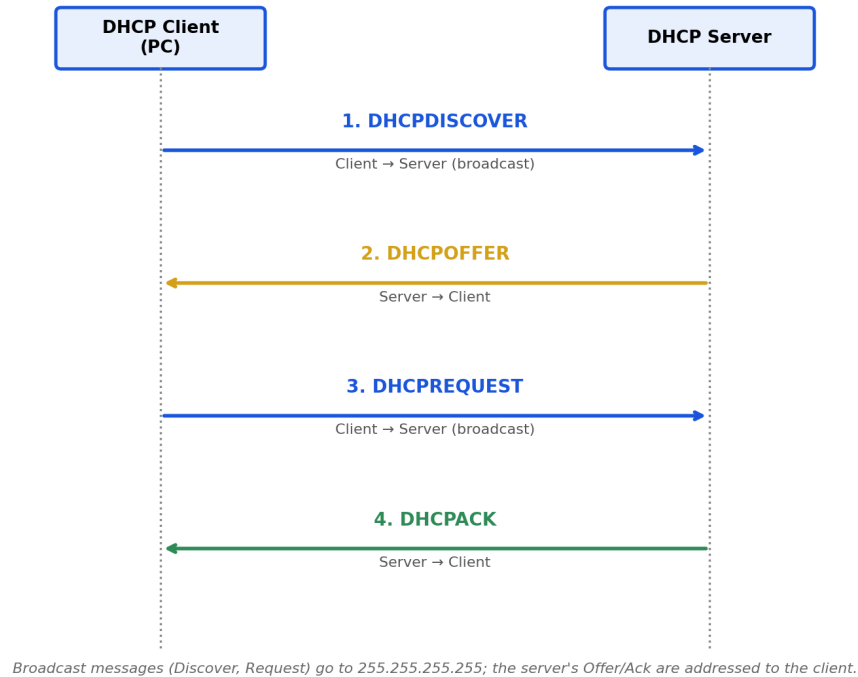
Packet Tracer Topology - DHCP Setup



- Built the topology above in Packet Tracer: DHCP Server (192.168.1.1) — Switch — PC0 and PC1. On the server I enabled the DHCP service, set the pool starting at 192.168.1.100 with the correct subnet mask and default gateway. On each PC's IP Configuration I selected DHCP instead of Static, and checked the assigned addresses — PC0 received 192.168.1.100 and PC1 received 192.168.1.101, confirming each client got a unique address from the pool.
2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.
Hint: Use arrows to show the direction of each message between client and server.

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DHCP Process (DORA)



3. **Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.**
 - Changed the pool so it only had one usable address (192.168.1.100) available. PC0 requested first and successfully received 192.168.1.100 through the normal DORA process. When PC1 then sent its DHCPDISCOVER, the server had no more addresses left in the pool to offer, so PC1's IP configuration stayed empty/showed "DHCP request failed" and it fell back to an APIPA-style unconfigured state.
 - This happens because a DHCP pool is a finite, shared set of addresses — once every address in the pool is leased out to a client, the server simply has nothing left to offer, so any further client asking for an address gets no response until an existing lease is released or expires.
4. **Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.**

Device	Assigned IP
DHCP Server (Restaurant Router)	192.168.5.1
Laptop (customer)	192.168.5.100
"Smartphone" (generic PC)	192.168.5.101

- DHCP is essential here because a restaurant's WiFi has customers constantly joining and leaving with different devices throughout the day. Nobody wants to manually type in an IP address, subnet mask, and gateway just to check Zomato over WiFi — DHCP lets every new device get connected automatically within seconds, and automatically frees up the address again once the customer leaves and their lease expires, so the same small pool of addresses can keep being reused for new customers all day.

Assignment : Network Protocols & Services

Day 3 – Assignment

1. **Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.**

→ Installed Simple DNS Plus on a Windows VM. In the console, right-clicked on “Forward Lookup Zones” and chose New Zone, then entered musicmania.in as the zone name and set it up as a Primary zone stored locally, which created the initial zone file with default SOA and NS records.

2. **Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.**

→ Added a new A record inside the musicmania.in zone: Host = www, IP address = 192.168.1.50, then saved and reloaded the zone.

```
C:\Users\Shah>nslookup www.musicmania.in 192.168.1.50
Server: localhost
Address: 192.168.1.50
```

```
Name: www.musicmania.in
Address: 192.168.1.50
```

→ The lookup returned exactly 192.168.1.50, confirming the A record is working correctly.

3. **Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.**

→ Added a CNAME record: Alias = play, points to = www.musicmania.in (the existing A record).

```
C:\Users\Shah>nslookup play.musicmania.in 192.168.1.50
Server: localhost
Address: 192.168.1.50
```

```
Name: www.musicmania.in
Address: 192.168.1.50
Aliases: play.musicmania.in
```

→ play.musicmania.in correctly resolved to 192.168.1.50 — the DNS server followed the CNAME alias to www.musicmania.in and then returned that record's A address, exactly as expected.

4. **Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.**
Hint: Include at least three different troubleshooting commands or checks in your documentation.

1. Check the zone file/configuration for musicmania.in for typos in record names, missing trailing dots, or a wrong IP address in the A record.
2. Run nslookup musicmania.in 127.0.0.1 (pointing directly at the local DNS server) to see whether the problem is the DNS server itself or something else, like the client's DNS settings.
3. Restart the DNS service (e.g. net stop/net start on the Simple DNS Plus service, or systemctl restart bind9 on Linux) in case the zone changes weren't reloaded.
4. Use dig musicmania.in +trace (or check the server logs) to see exactly where resolution is failing — e.g. no response at all vs. NXDOMAIN vs. an old cached answer.
5. Check that the firewall on the DNS server isn't blocking port 53 (both TCP and UDP), since that would stop nslookup/dig queries from ever reaching the service.

Assignment : Network Protocols & Services

Day 4 – Assignment

- 1. Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.**
 - Downloaded and installed FileZilla Server, then opened the FileZilla Server Interface and connected to the local server.
 - Went to Edit → Users, clicked Add, and created a user account “shahftp” with a password.
 - Under Shared folders for that user, added C:\FTP_Share as the home directory and gave it Read/Write/Delete permissions.
 - Made sure the server was listening on port 21 and that Windows Firewall allowed FileZilla Server through.
- 2. Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.**
 - Opened FileZilla Client, entered Host: 127.0.0.1, Username: shahftp, Password: (the one set earlier), Port: 21, and clicked Quickconnect. Once connected, I dragged a photo (sunset.jpg) from the local file panel on the left into the remote FTP_Share folder on the right, and it uploaded successfully, appearing in the remote file listing.
- 3. Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.**
 - In FileZilla Client, navigated the local panel to a new folder (Downloads\FTP_Test), then right-clicked sunset.jpg in the remote panel and selected Download. Checked the file size of both copies in File Explorer — the original was 2.4 MB and the downloaded copy was also exactly 2.4 MB, confirming the transfer completed without corruption.
- 4. Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).
Hint: Think about how network admins monitor device health, bandwidth, or uptime.**
 - An ISP (like Jio or Airtel) uses SNMP to remotely monitor thousands of routers and modems across a city — checking their uptime, signal strength, and bandwidth usage from a central management dashboard, so they can spot and fix outages before too many customers complain.
 - A college's campus WiFi network uses SNMP to monitor all the WiFi access points spread across buildings — tracking how many devices are connected to each AP and their current load, so the IT team can tell if one access point is overloaded and needs load balancing or a firmware update.
 - A large website's data centre uses SNMP to monitor the health of its switches and load balancers — collecting metrics like CPU temperature, interface traffic, and error rates, feeding into monitoring dashboards that alert engineers automatically if a device starts failing.
- 5. Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.**

Point	FTP	HTTP
Port Number	21 (control), 20 (data)	80 (HTTP), 443 (HTTPS)
Security	Plain text by default (FTPS/SFTP add encryption)	Plain text (HTTP) or encrypted (HTTPS)
Data Transfer Method	Uses separate control and data connections	Single connection carries both request and response

Assignment : Network Protocols & Services

Point	FTP	HTTP
Use Cases	Bulk file uploads/downloads, website deployment	Browsing websites, APIs, streaming, web apps
Authentication	Requires a username and password to log in	Often no login needed; auth added at the app level (cookies, tokens) when required

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Day 5 – Assignment

1. **Simulate a DNS resolution failure on your local machine by editing the hosts file to point `www.spotify.com` to a wrong IP address, then try to ping the domain and document the error message you receive.**
 - Opened `C:\Windows\System32\drivers\etc\hosts` as administrator and added the line: `10.10.10.10 www.spotify.com` — an address that doesn't actually host Spotify.

```
C:\Users\Shah>ping www.spotify.com

Pinging www.spotify.com [10.10.10.10] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```
 - The ping resolved the name to the wrong IP (10.10.10.10) exactly as I'd set in the hosts file, then failed with “Request timed out” for every packet since nothing is actually listening at that address — showing how a bad hosts file entry can silently break a working domain even though DNS itself is fine.
2. **Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.**
 - On the DHCP server's pool settings, I deliberately set the default gateway to 192.168.1.254 instead of the router's real address, 192.168.1.1. After connecting a PC and letting it pull an address via DHCP, its IP Configuration showed Default Gateway: 192.168.1.254 exactly as configured, and the PC could reach other devices on the same LAN but couldn't reach anything outside it (like the simulated internet).
 - To identify this, I'd run `ipconfig` on the client and compare the gateway shown against the router's actual LAN interface IP — a mismatch here is the giveaway. To fix it, I'd go back into the DHCP server's pool configuration, correct the default-gateway value to 192.168.1.1, and then have the client release/renew its lease (or simply reconnect) to pick up the corrected gateway.
3. **Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the `nslookup` or `dig` tool to diagnose the DNS problem and write down the troubleshooting steps and solution.**
 - Confirmed the symptom: ping 8.8.8.8 (by IP) worked fine, but ping `www.google.com` (by name) failed — pointing specifically at DNS rather than general connectivity.
 - Ran `nslookup www.google.com` and got “DNS request timed out” / no response, meaning the client couldn't reach its configured DNS server at all.
 - Ran `ipconfig /all` and checked the DNS Servers entry — found it was pointing to an old/incorrect DNS server IP that was no longer reachable.
 - Solution: updated the client's DNS server setting to a working DNS server (the local DNS server's correct IP, or a public one like 8.8.8.8), then re-ran `nslookup www.google.com` to confirm it now returned a valid IP.
4. **A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the**

Assignment : Network Protocols & Services

step-by-step process you would use to resolve it.
Hint:
Check for network connectivity, DHCP scope exhaustion, and server availability.

- Possible cause 1: No physical/logical connectivity between the client and the DHCP server (bad cable, wrong VLAN, or a switch port that's down), so the DHCPDISCOVER broadcast never reaches the server.
 - Possible cause 2: The DHCP scope/pool is exhausted — every address is already leased out, so the server has nothing left to offer.
 - Possible cause 3: The DHCP server itself is down, its service has stopped, or the scope was never activated.
1. Check the physical/logical connection first — confirm the cable, port, and VLAN are correct.
 2. On the DHCP server, check whether the scope is active and how many addresses remain available in the pool.
 3. Confirm the DHCP service itself is running (restart it if needed).
 4. Once the cause is fixed, run `ipconfig /release` followed by `ipconfig /renew` on the client to request a fresh lease and confirm it now receives a proper address.
5. **Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.**

Issue	Symptom	Possible Cause	Solution
DNS – wrong hosts entry	Wrong site loads, or ping goes to the wrong IP	Incorrect entry in the local hosts file	Remove/correct the bad line in the hosts file
DNS – server unreachable	“DNS request timed out” on nslookup	Configured DNS server IP is wrong or offline	Point the client to a working/correct DNS server
DNS – stale cache	Old IP still resolves after a record was changed	Local DNS cache hasn't expired yet	Run <code>ipconfig /flushdns</code> to clear the cache
DHCP – scope exhausted	New device gets a 169.x.x.x APIPA address	All addresses in the DHCP pool are leased out	Free up unused leases or expand the DHCP pool
DHCP – wrong gateway/DNS handed out	Client connects to LAN but can't reach the internet	DHCP scope options misconfigured	Correct the gateway/DNS values in the DHCP scope
DHCP – service down	No client on the network gets an IP at all	DHCP server service stopped or scope not activated	Restart the DHCP service and activate the scope

Assignment : Network Protocols & Services

Days 1–5 – Final Project: Build and Configure a Small Office Network

1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Device	Static IP	Subnet Mask
PC-1	192.168.10.10	255.255.255.0
PC-2	192.168.10.11	255.255.255.0
PC-3	192.168.10.12	255.255.255.0
PC-4	192.168.10.13	255.255.255.0
PC-5	192.168.10.14	255.255.255.0

```
C:\Users\Shah>ping 192.168.10.11
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

- Pinged between each pair of the 5 devices and all replied successfully with 0% packet loss, confirming they can all communicate within the same subnet.

2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Device	Assigned IP (DHCP)
PC-6	192.168.10.100
PC-7	192.168.10.101
Smart TV	192.168.10.102

- Each of the 3 devices received a different address from the 192.168.10.100–.200 DHCP pool, confirmed by checking ipconfig on each device.

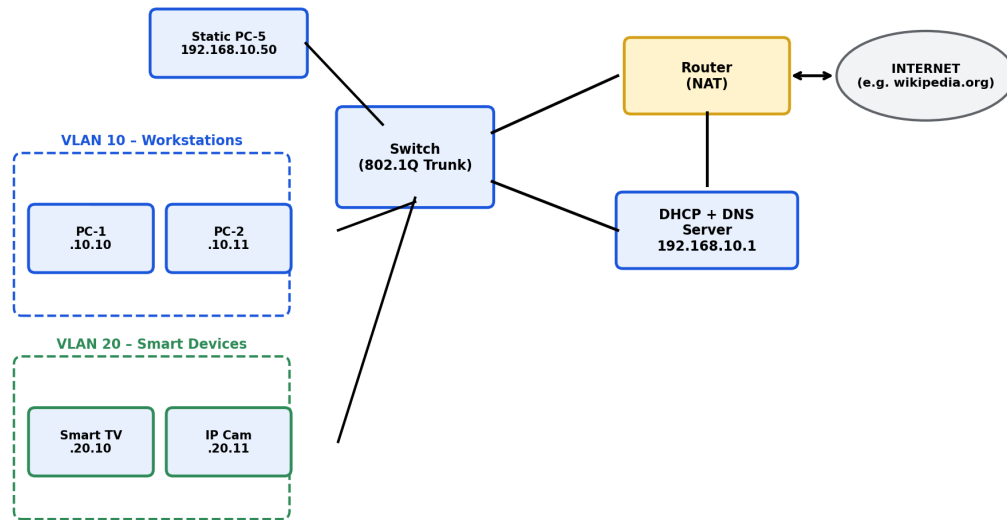
3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.
Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

- Set up dnsmasq on the network's Linux gateway box and added the line:
address=/musicplayer.local/192.168.10.20 to its configuration, then set dnsmasq as the DNS server pushed out by DHCP to all clients. After that, typing http://musicplayer.local in a browser on any device on the network correctly opened the local media server's web interface running at 192.168.10.20, without needing to remember or type the raw IP address.

4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.
Hint: Use ping or file sharing to test connectivity between devices in different VLANs.

Assignment : Network Protocols & Services

Small Office Network - Final Topology



- Created VLAN 10 (“Workstations”) and VLAN 20 (“Smart Devices”) on the switch, then assigned the switch ports for PC-1/PC-2 to VLAN 10 and the ports for the Smart TV/IP Camera to VLAN 20.
- Ping test within VLAN 10: PC-1 (192.168.10.10) successfully pinged PC-2 (192.168.10.11) — 0% packet loss.
- Ping test across VLANs: PC-1 (VLAN 10) tried to ping the Smart TV (VLAN 20) and got “Request timed out” for every packet, since the switch keeps the two VLANs logically separated as if they were on different physical networks, even though they share the same physical switch.

5. **Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.**

- Configured the router's WAN interface with NAT (ip nat outside / ip nat inside on the LAN-facing interface in Packet Tracer, along with an “ip nat inside source list ... interface ... overload” rule), so that all the private 192.168.10.x addresses would be translated to the router's single public-facing IP when leaving the network.

```
C:\Users\Shah>ping www.wikipedia.org
Pinging www.wikipedia.org [198.35.26.96] with 32 bytes of data:
Reply from 198.35.26.96: bytes=32 time=32ms TTL=54
Reply from 198.35.26.96: bytes=32 time=30ms TTL=54
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```
- The ping to the external site succeeded with 0% loss, confirming that devices on the private LAN can now reach the outside internet through the router's NAT translation, even though they only have private (non-routable) IP addresses internally.